

Startup Rebuild — Design Brief

Source of truth for `STARTUP-REBUILD-CHECKLIST.md` . Section numbers (S1–S16) are the refs used by every task line in that checklist. Nothing here is implemented yet; the removal checklist is the first work.

S1. Purpose and scope

The entire startup surface is deleted and rebuilt from scratch: the `MTM_Waitlist.Startup` project, the `Module_Startup` views, the launch orchestration in `App.xaml.cs` , the activation surface, the startup contracts, models and options in `MTM_Waitlist.Core` , startup logging, the startup tests, the capabilities/startup living spec and its registry entry, and the docs that describe the old splash.

The old logic is replaced first by a **placeholder window** so the removal is provable: if the app can no longer reach the shell, the old logic is gone. Only then is the new module built.

S2. Decisions locked (2026-09-25)

#	Decision
1	Rebuild <code>MTM_Waitlist.Startup</code> in place , same name; views stay in <code>Module_Startup/Views</code> .
2	Old material is deleted outright — git history is the archive. No <code>_legacy/</code> parking.
3	Sign-in is rebuilt too; it counts as startup.
4	The placeholder splash is splash only — the app stops there and nothing else loads.
5	Steps become data with a derived count, and the splash shows everything the process is doing .
6	Errors surface as a toast anchored to the bottom of the splash window .
7	The session moves into the store. Only connection strings may remain local .
8	The logging destination stays a hard gate , satisfied by a new machine configuration step.

#	Decision
9	Machine configuration is reachable before any operator sign-in , unlocked by an IT Department or Developer sign-in.
10	All logging goes to the database ; logging becomes its own module with a developer log panel .
11	One blocked-state surface; Restore Defaults states exactly what it will reset and resets only what is broken .
12	Where a reset can be done silently without hurting the operator, do it silently .
13	Roles come from the store only — the <code>appsettings.json</code> developer allow-list is gone.
14	<code>StartupState</code> is replaced by two narrow contracts : person identity, and machine facts.
15	Startup keeps: the picture-cache refresh, the Infor Visual verdict priming, the forced password change, and the computer-registration gate.
16	The log-folder prompt is gone entirely.
17	Remember-me survives, stored in the store against the person, encrypted.
18	The encryption key is read from a shared key file (S8.3).
19	The session lives in a new <code>user_active_sessions</code> table.
20	Every deployed table, procedure, function and view is audited for dead weight; dead SQL files are deleted.

S3. What is being removed

S3.1 Code, tests, docs

- `MTM_Waitlist.Startup/**` (all services, view models, DI extension, csproj) — including the `sln` entry, the app's `ProjectReference` and `Compile Remove`, and the test project's `ProjectReference`.
- `Module_Startup/Views/**` (`SplashWindow`, `SplashView`, `SplashPage`, `LoginWindow`, `LoginPage`).
- `App.xaml.cs`: the splash/login window statics, `ShowSplashWindow`, `ShowMainWindowAndCloseSplash`, `ShowLoginWindowAndCloseSplash`, `ShowMainWindowAndCloseLoginWindow`, `CloseSplashWindow`, and the three window-closed handlers.

- `MTM_Waitlist.Core` : `ActivationService` , `Activation/**` , every `IStartup*` contract, `StartupState` , `StartupResult` , `StartupSessionSnapshot` , `StartupCredentialCheckResult` , `StartupPasswordResetRequirement` , `IComputerGateService` , `IComputerRegistryService` , `ISignOutService` , `SignOutResult` , `IAppProcessRestarter` , `IStartupWindowService` , `IStartupShellStateService` , `IStartupLogService` , `IStartupLogForwarder` , and the four `Startup*Options` models.
- `Services/AppLifecycleService.cs` and the `IAppLifecycleService` seam it implements.
- `StartupDebugLog` — see S9 for its 491 call sites.
- All startup tests, including `SplashViewModelTests` , `LoginViewModelTests` , `StartupCoordinatorTests` , `StartupLogServiceTests` , `StartupRecoveryServiceTests` , `ComputerGateServiceTests` , `ComputerRegistryServiceTests` , `TemporaryCredentialLimitTests` , `StartupArchiveCleanupTests` , `SignOutServiceTests` , `StartupModelsTests` , `StartupOptionsModelsTests` , `StartupDebugLogTests` , and `NoOpAppLifecycleService` .
- `capabilities/startup/**` , the startup entry in `living-specs.yml` , and the startup row in `capabilities/DRIFT.md` .
- `STARTUP-FLOW.md` , `STARTUP-FLOW.html` , `STARTUP-FLOW.pdf` ; the splash/sign-in steps in `.github/instructions/winui3-ui-automation.instructions.md` ; the startup mentions in `.github/instructions/test-conventions.instructions.md` ; the module list in `README.md` .
- The eight `Shell_SignOut.*` / `Startup_SignIn.*` resource keys and the fourteen `Startup_*` tooltip keys (in **both** `TooltipResources.resw` and `TooltipResources.developer.resw`).

S3.2 Shared artifacts that must survive untouched

Tables: `core_users_profiles` , `core_computers_registry` (three foreign keys point at it), `auth_roles_catalog` , `auth_roles_assignments` , `config_settings_values` .

Procedures: `sp_auth_user_row_get` (also serves the cache-service API),

`sp_auth_temporary_credential_attempt_record`

(also user management),

`sp_core_computers_registry_upsert|update|delete|get_all|registered_get|lookup_by_name_get`

(Settings and Work Center Catalogue), `sp_auth_roles_list` , `sp_config_settings_values_get|upsert` .

Seeds and aggregates: `seed_dev_masked_baseline` , `seed_username_upper_normalization` ,

`seed_role_admin_to_it_department` , `AllSeeds.sql` , `update_table_descriptions.sql` , `AllTables.sql` ,

`AllSPs.sql` , `AllFunct.sql` .

Core services: MySqlHostFallback , ILocalSettingsService , IFileService , MySqlHelperServer ,
fn_server_utc_now .

S3.3 Startup-only database artifacts (removable in isolation)

sp_server_utc_now_get , sp_auth_credentials_check , sp_auth_user_password_update ,
sp_auth_computer_registered_get , sp_auth_session_expiry_get ,
sp_auth_password_reset_required_get ,
sp_core_computers_registry_lookup_by_name_mac_get ,
sp_core_computers_registry_lookup_by_mac_get ,
sp_core_computers_registry_update_by_mac , and Database/Validation/startup_schema/ .
fn_server_utc_now is **retained** — the session rule in S8 depends on it.

S4. Target architecture

- MTM_Waitlist.Startup rebuilt as the launch pipeline library; XAML stays in Module_Startup/Views .
- The app host keeps a thin seam so App.xaml.cs does not grow a second orchestration layer: the new module
exposes one entry point that the host starts, and one event the host observes when the shell is ready.
- IAppLifecycleService is replaced by that seam; the host no longer owns window handoff.
- The shell is reached only through the new pipeline. There is no bypass and no "continue anyway" path.

S5. Splash and the activity feed

- The splash is one surface. It renders a **live, append-only activity feed**: one line per thing the process
does, with a timestamp, so a lock-up is attributable to a named line rather than to a step number.
- Progress is still announced **before** each check runs.
- An **error toast** is anchored to the bottom of the splash window. It carries the plain-language diagnosis
and never covers the feed.
- The feed is the debugging surface for a live fault: it must name the operation, its target (store, share,
server) and its outcome.
- Window sizing stays configurable and the sizing failure path stays non-fatal.

S6. The step model

- Steps are **data**, not a hard-coded five: each step has a name, a description and a category.
- The displayed count is **derived** from the step list, so adding or removing a step never leaves a stale "of 5".
- The feed shows started / completed / failed transitions per step, plus sub-operations inside a step.
- Retry re-runs the failed step and the steps after it — never the whole pipeline, and never step 1 for a database-only retry.

S7. Identity contracts

Two interfaces replace `StartupState` :

1. **Person identity** — read-only: user id, sign-in name, display name, employee number, current role code, and the role codes the person holds. No writer other than the pipeline.
2. **Machine facts** — read-only: normalised hostname, normalised MAC address, the registered computer row (id, display name, description), and whether this machine is registered.

Consumers to migrate (~20): permissions, user management, waitlist attribution, tooltips and the control

inspector, Settings computer screens, picture preferences, and the shell's user badge.

`PermissionGateSiteAuditTests` currently pins `MTM_Waitlist.Core/Models/StartupState.cs` and its `IsDeveloper` body; it is rewritten against the new contracts.

S8. Session, remember-me, and secret material

S8.1 Session

- The session lives in a new `user_active_sessions` table: one active row per person per machine, with the token stored as a **salted hash**, an expiry, an issued time, and the machine key.
- Session validity is judged on the **store's clock** (`fn_server_utc_now`), never the workstation's.

- The app writes the row when sign-in completes (only after the machine gate passes) and clears it on sign-out.
- The eight-hour lifetime is retained unless the user says otherwise.

S8.2 Remember-me

- Stored in the store against the person, never in a local file.
- **Encrypted, not hashed** — it must be decryptable. See S8.3 for the key.
- A failed decrypt falls back to the normal sign-in form and records the fault; it never blocks startup.

S8.3 The key file

- Key path:
`X:\Software Development\Live Applications\MTM_Application_Keys\MTM_AUTH_USER_SECRET_KEY.txt`
- **Implementation note:** x: is a mapped drive letter and mapped drives are per-user session state. The implementation must resolve and store the UNC equivalent, matching the existing key-folder precedent
`( ImageStorage.KeysFolderPath )`. Confirm the UNC form before coding.
- The key file is never logged, never copied into the repository, and never echoed by the developer log panel.
- A Security Review task covers key handling, and a Security Engineer persona owns it.

S8.4 Local state that may remain

Only the connection strings used to reach the store. Everything else — theme, remember-me, waitlist sort

order, ignored locations, new-request alerts, waitlist message-seen — moves into the store. A double pass

(S11.4) must prove nothing else is left.

S9. Logging

- Logging becomes **its own module**, separate from startup.
- Every entry goes to the **database**. There are no local log files and no log-folder prompt.
- `ops_startup_logs` already carries `correlation_id`, `previous_hash` and `entry_hash` and has no writer

today; it (or a renamed successor) becomes the store, with create/rollback procedures, an index set for the panel's filters, and a retention routine.

- Every entry carries: timestamp, severity, area/module, machine, user, error type, message, exception detail, and the hash chain link.
- StartupDebugLog 's 491 call sites across 85 files survive behind a new seam with the same call ergonomics.
- A **developer log panel** in Settings lists and filters by criticality, machine, user, module and error type.
- When the store is unreachable the app refuses to start, so nothing is persisted for that case — the splash shows the diagnosis and the operator reports it (decision 12 of S2).

S10. New machine configuration

- A **machine-setup screen** appears when the machine is not configured, reachable **before any operator sign-in**.
- It is unlocked by an **IT Department or Developer** sign-in, which authorises configuration only and does not open the shell.
- It captures what the old local settings held and the destination needs: the machine's display name and description, the shared picture sources, and the logging destination.
- The privilege surface must be updated in the same change: new permission keys in the permission catalogue, the Settings privileges page, the role baselines, and the seeds. The user expects the Settings page to break without this work.

S11. Database changes

S11.1 New

- `user_active_sessions` — table, create + rollback, procedures for write/validate/clear, master-list registration, and an entry in a validation script.

S11.2 Repurposed

- The log store (S9) gains its writer, its reader procedures, its retention routine and its indexes.

S11.3 Dead-weight audit

- Every deployed table, procedure, function and view is checked for any consumer. Artifacts with none have their SQL files **deleted** — `auth_sessions_tokens` is the first candidate, and `ops_startup_logs` is repurposed rather than deleted. Candidates already known: `core_buildings_catalog` , `core_buildings_history` , `sp_core_buildings_upsert` .
- Each deletion updates the aggregates (`AllTables.sql` , `AllSPs.sql` , `AllFunct.sql` , `AllViews.sql` , `AllSeeds.sql`), the validation scripts, and any doc that names the artifact.
- A SHARED artifact must never be deleted by this audit (S3.2).

S11.4 The local-settings double pass

- Pass 1: every `ILocalSettingsService` call site, every `LocalSettings.json` key in a live file, and every `ApplicationData / ApplicationDataContainer` use.
- Pass 2: every `appsettings.json` section read by the app, and every file the app writes outside the image cache.
- Result recorded in this brief, with the remaining local surface reduced to connection strings and a task per remaining key to move or delete it.

S12. Failure semantics

- A blocking condition states a diagnosis specific to its cause; a blocked startup never reaches the shell.
- **Restore Defaults** first shows a popup listing exactly what will be reset, then resets only what is broken — this machine's store-side configuration rows. It never touches people, roles or permissions.
- Where a fault can be repaired silently without hurting the operator, it is repaired silently (decision 12).

- Retry repeats the failed step; a database fault offers no reset that cannot repair it.
- Unreachable picture sources and unreachable Infor Visual remain **best effort** and never block the start.
- The picture-cache step and the Infor Visual priming each get a **bounded** wait; the two unbounded waits found in the sweep (an unloaded splash view, and a stalled share with no cancellation) must not be re-created.

S13. Edge cases to preserve

The 111 cases found by the three sweeps are the acceptance surface for the rebuild. Grouped:

1. Settings/config guards and targeted repair (6).
2. Connection string, store reads, session precedence and clock rules (13).
3. Identity, role resolution and the (now removed) developer override (3 retained, 2 dropped).
4. Password and temporary-credential rules, including the five-attempt limit (13).
5. Computer gate and registration outcomes, including the duplicate-name and no-MAC paths (13).
6. Window, splash and shell-state behaviour, including maximize fallback (14).
7. Blocked-state remedies, now with the targeted reset (9).
8. Picture-cache best-effort behaviour and archive retention (11).
9. Diagnostic guarantees: never waits, never fails, survives shutdown, bounded retention (12).
10. Reachability probing and hysteresis (5).
11. Sign-out and relaunch, including a refused relaunch (6).
12. Activation, notifications and deep-link routing (4).

Cases that **change** under these decisions: the developer allow-list override, the log-folder prompt, local

session precedence, local-file repair, and Restore-Defaults-offers (now always offered, but scoped).

S14. Constraints and non-goals

- No self-service password reset; no reset offered on the sign-in screen.
- A reset issues a one-time four-digit PIN; only a salted hash is stored; the legacy 0000 stays recognised.
- A reset must not end an existing session.
- The attempt limit applies only to accounts holding a temporary credential; ordinary sign-in has no limit.
- No "continue anyway" past the startup gate, no optional connection check, no dismissible banner.

- No manual demo/mock mode, ever.
- Plaintext credentials and tokens are never stored; tokens are held as salted hashes.
- Every schema artifact ships `create.sql` + `rollback.sql` and is registered in the aggregate files.

S15. Risks and known conflicts

Risk	Why it matters
x: in the key path is a mapped drive, not a share	Will not resolve for another account, session or service. Must be stored as UNC (S8.3).
Roles from the store only	Kills the no-sign-in developer walkthrough that <code>READINESS-CHECKLIST.md</code> and the UI-automation instructions rely on.
"Nothing local" touches six unrelated features	Theme, sort order, ignored locations, alerts and message-seen all currently live in <code>LocalSettings.json</code> .
Logging to the store cannot record a store outage	Accepted in S9; the splash is the only record for that case.
Two audit tests pin the old shapes	<code>PermissionGateSiteAuditTests</code> and <code>PageActivationAuditTests</code> must be rewritten in the same change.
The corrupt-settings repair path loses its cause	It existed because settings were a local JSON file; most of its tests become meaningless.

S16. Verification

- The retired-symbol audit fails the build if any removed name returns.
- After the removal and placeholder, the app must not reach the shell by any path.
- The suite's startup tests are deleted, not disabled; a green run afterwards must be attributable to the new tests alone.
- The database schema validator runs clean after the dead-weight removals.
- Each reconstructed edge case in S13 gets a test in the new module, or a written reason why it is gone.